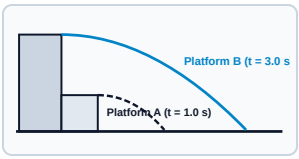
Honors Physics: CAST Projectile Motion Data Analysis

UNIT 2 WRAP-UP • DATA TABLES, GRAPH SLOPES, TRIG COMPONENTS & CER

Name: _____ **Per:** _____ **Date:** _____

EXPERIMENTAL SETUP: TWO-PART PROJECTILE INVESTIGATION

Gravity: g = 10 m/s²



- **Investigation 1 (Horizontal Launches):** Balls are fired horizontally (θ = **0°**) from **Platform A** (t = **1.0 s**) and **Platform B** (t = **3.0 s**) across speeds **0–20 m/s**.
- **Investigation 2 (Angled Ground Launches):** On level ground at constant v_o = **20 m/s**, students test five launch angles (**15°–75°**). The **60° row** in Table 2 has missing values that you will calculate in Part 2.

✏️ **Scan Card Instructions (Right):** Write & bubble your **6-digit Student ID** clearly inside the box on the right. **Leave the SCORE bubbles blank** for your teacher to bubble & scan!

CAST Projectile Data Analysis **MAX SCORE: 10**

Unit 2 Honors Telemetry Synthesis

Bubble your 6-digit Student ID. Leave SCORE blank for teacher.

STUDENT ID

☐	☐	☐	☐	☐	☐
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

SCORE

☐	☐	/ 10
0	0	
1	1	
2	2	
3	3	
4	4	
5	5	
6	6	
7	7	
8	8	
9	9	

THE GRADEST OMR

INVESTIGATION 1: HORIZONTAL LAUNCH SPEED (V_{OX}) VS. LANDING RANGE (ΔX)

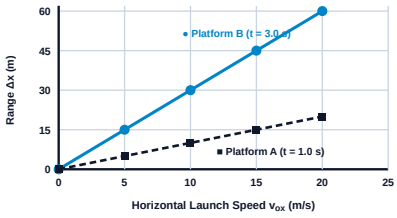
Platform A (t = 1.0 s) vs. Platform B (t = 3.0 s)

Table 1: Horizontal Launch Range from Two Platforms (θ = 0°)

Horizontal Speed v_{ox} (m/s)	Platform A (t = 1.0 s) Range Δx_A (m) [■]	Platform B (t = 3.0 s) Range Δx_B (m) [●]
0	0	0
5	5	15
10	10	30
15	15	45
20	20	60

📐 **Physics Connection:** Because $\Delta x = v_{ox} \times t$, on a graph of **Range (Δx)** vs. **Horizontal Speed (v_{ox})**, the **slope ($\Delta x \div v_{ox}$)** equals the projectile's **flight time (t)** in seconds!

Graph 1: Landing Range (Δx) vs. Horizontal Speed (v_{ox})



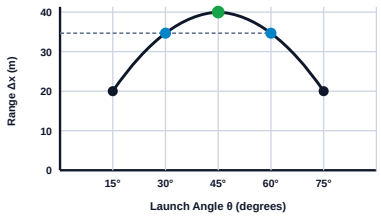
INVESTIGATION 2: LAUNCH ANGLE (Θ) VS. VELOCITY COMPONENTS, HANG TIME & RANGE

v_o = 20 m/s | $v_{ox} = v_o \cos(\theta)$ | $v_{oy} = v_o \sin(\theta)$

Table 2: Angled Ground Launch Data (v_o = 20 m/s)

Angle θ (°)	Horiz. Speed v_{ox} (m/s)	Initial Vert. Speed v_{oy} (m/s)	Hang Time t_{hang} (s)	Landing Range Δx (m)
15°	19.3	5.2	1.04	20.0
30°	17.3	10.0	2.00	34.6
45°	14.1	14.1	2.83	40.0
60°	? (Part 2)	? (Part 2)	3.46	? (Part 2)
75°	5.2	19.3	3.86	20.0

Graph 2: Landing Range (Δx) vs. Launch Angle (θ)



 Formulas: $v_{ox} = v_o \cos(\theta)$ | $v_{oy} = v_o \sin(\theta)$ | $\Delta x = v_{ox} \times t_{hang}$

PART 1: READING MULTI-SERIES GRAPHS & CALCULATING SLOPE (3 POINTS)

Investigation 1 & 2

☐ **(1A) READ & CLASSIFY**

1. On **Graph 1** at v_{ox} = **15 m/s**: Platform A Range (Δx_A) = _____ **m** | Platform B Range (Δx_B) = _____ **m**.

2. Circle the trend: **Graph 1** shows a **[linear direct relationship / non-linear arch]**, while **Graph 2** shows a **[linear direct relationship / non-linear symmetric arch]** peaking at _____ °.

☐ **(1B) CALCULATE SLOPE OF PLATFORM B LINE (•)**

From v_{ox} = 0 to 20 m/s: Slope = $\Delta x_B \div v_{ox}$

Slope = _____ m ÷ _____ m/s = _____ s (Physical Meaning: _____)

☐ **(1C) EXTRAPOLATE RANGE AT V_{OX} = 25 M/S**

Predict Platform B range (t = 3.0 s) at v_{ox} = 25 m/s:

$\Delta x_B = v_{ox} \times t$ = _____ m/s × _____ s = _____ m

PART 2: COMPLETING MISSING 60° TABLE DATA USING TRIG & RANGE FORMULA (3 POINTS)

Given: cos(60°) = 0.50 | sin(60°) = 0.866

☐ **(2A) FIND VELOCITY COMPONENTS AT Θ = 60° (V_O = 20 M/S)**

1. Horizontal Speed ($v_{ox} = v_o \cos 60°$):

v_{ox} = 20 m/s × 0.50 = _____ m/s

2. Initial Vertical Speed ($v_{oy} = v_o \sin 60°$):

v_{oy} = 20 m/s × 0.866 = _____ m/s

☐ **(2B) CALCULATE MISSING LANDING RANGE AT Θ = 60°**

Use your v_{ox} from (2A) and t_{hang} = 3.46 s from Table 2:

$\Delta x = v_{ox} \times t_{hang}$

Δx = _____ m/s × 3.46 s = _____ m

Which other launch angle in Table 2 lands at this exact same distance? _____ °

PART 3: CAST CONSTRUCTED RESPONSE — CLAIM, EVIDENCE & REASONING (4 POINTS)

Rubric: Claim (1) + Evidence (2) + Reasoning (1)

🧪 **CAST Investigation Prompt:** During Investigation 2, a student argues: *"A launch angle of 30° and a launch angle of 60° cannot possibly land at the same distance because the 60° shot stays in the air much longer."*

Evaluate the student's argument using quantitative evidence from Table 2 (including your 60° calculations) and $\Delta x = v_{ox} \times t_{hang}$.

💬 **Optional CAST Sentence Starters:** • ***Claim:*** *"The student's argument is incorrect because 30° and 60° are complementary angles that both land at..."*

• ***Evidence:*** *"According to Table 2, at 30° the horizontal speed is ___ m/s and hang time is ___ s (Range = ___ m), whereas at 60°..."*

• ***Reasoning:*** *"Because horizontal range equals $v_{ox} \times t_{hang}$, switching from 30° to 60° trades horizontal speed for hang time..."*

☐ **ZONE A — CLAIM (1 pt): State whether 30° and 60° land at the same distance and identify that distance.**

☐ **ZONE B — EVIDENCE (2 pts): Cite specific numbers & units from Table 2 for BOTH 30° and 60° (v_{ox}, v_{oy}, t_{hang}, and Δx).**

☐ **ZONE C — REASONING (1 pt): Explain WHY multiplying $v_{ox} \times t_{hang}$ gives the same range for 30° and 60°, and state why 45° gives the maximum range.**